

Chapter 4

FORCED CONVECTION

The concept and definitions of heat transfer coefficients, individual and overall, have been introduced in Chapter 3. We explained with examples how simple problems involving convective transport of heat can be solved if the *heat transfer coefficient* is known. But the question remains as to how we can obtain the heat transfer coefficient in a given situation. The heat transfer coefficient is not as simple a quantity as a fundamental property of a body or a system. It depends upon many factors—the physical properties of the fluid (like thermal conductivity, density, viscosity), the velocity field, the geometry of the system and its characteristic dimensions, etc. Fortunately, a large number of empirical correlations have been developed over the years by heat transfer researchers. These correlations cover a great majority of situations encountered in industrial practice, and are used for the calculation of heat transfer coefficients. No doubt, the use of any correlation involves some error—that may be even 25–30% in some cases. However, in the absence of more reliable data or information in specific cases, these correlations are indispensable for the estimation of heat transfer coefficients.

In this chapter we will first qualitatively analyze convective heat transfer in a few simple systems in order to provide further understanding of this important mode of heat transfer. Later, we will discuss the more important heat transfer correlations and illustrate their uses. Convective heat transfer may be of two types *forced convection* and *free* or *natural convection*. Both are caused by motion in the medium. If the motion or velocity in the medium is generated by the application of an external force (e.g. by a pump, a blower, an agitator, etc.), heat transfer is said to occur by forced convection. But if the motion in the medium occurs as a result of density difference (which may be caused by a temperature difference), the concerned mode of heat transfer is called free convection. In this chapter we will confine our attention to forced convection. Free convection will be discussed in Chapter 5.

4.1. FORCED CONVECTION IN SYSTEMS OF SIMPLE GEOMETRIES

Here we identify a few simple flow situations for understanding the physical mechanism of forced convection heat transfer, without going into the mathematical analysis. Although the simplest way to visualize convective heat transfer to or from a surface is to assume the existence of a *stagnant* fluid film at the wall, *no* such film does exist in reality. It is true, however, that in the case of heat transfer from a surface, most of the resistance to heat transfer is offered by a narrow zone of fluid near the surface, called the boundary layer, but the fluid in this zone is never stagnant. The concept of boundary layer is of vital importance in understanding the flow and heat transfer characteristics near a wall.

4.1.1. Flow over a Flat Plate

Let us consider the flow of a fluid over an immersed wide flat plate [Fig. 4.1(a)] at zero angle of incidence (which means that the plate is oriented along the direction of flow of the bulk fluid).

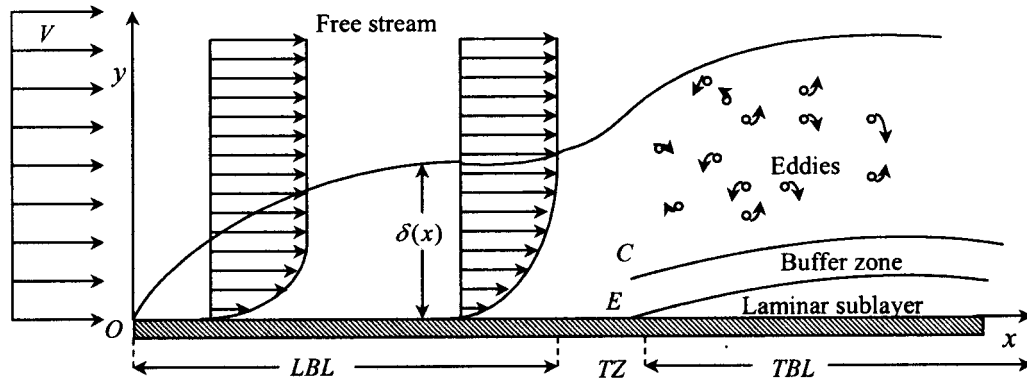


Fig. 4.1 (a) Boundary layer over a flat plate (at zero angle of incidence). (LBL: laminar boundary layer, TZ: transition zone, TBL: turbulent boundary layer.)

Fluid velocity at the surface of the plate is zero (because of no-slip condition), and gradually increases with distance from the plate. At a sufficiently large distance from the plate, the fluid velocity becomes equal to the 'free stream velocity', V . The region above the plate surface within which this change of velocity from zero to the free stream value occurs is called the *boundary layer*, or more specifically, the *velocity boundary layer* (also called the *momentum boundary layer* or the *hydrodynamic boundary layer*). The thickness of this region is called the *boundary layer thickness* and is denoted by δ . The boundary layer thickness increases with the distance x from the leading edge of the plate, i.e $\delta = \delta(x)$.

A theoretical analysis of boundary layer flow indicates that the transverse distance from the plate where the free stream velocity is attained is very large. As a result, the thickness of the boundary layer is rather arbitrarily defined as the distance from the plate at which a certain percentage, usually 99%, of the free stream velocity is attained.

The characteristics of boundary layer flow depend upon the Reynolds number defined as $Re_x = \rho V x / \mu$, where x is the distance from the point O or the leading edge. Because x is a variable, Re_x is called the 'local Reynolds number'. The fluid motion in the boundary layer remains laminar up to a point on the plate where $10^5 < Re_x < 5 \times 10^5$, which is the *critical Reynolds number* (it depends upon the roughness of the plate) for boundary layer flow. When Re_x increases beyond this value, transition to turbulent boundary layer begins. At $Re_x = 10^6$, the boundary layer becomes fully turbulent. However, these values should not be taken too strictly because they depend upon several factors including the surface roughness of the plate. The laminar, transition as well as the turbulent portions of the boundary layer are shown in Fig. 4.1(a). The velocity profile in the laminar region is also shown.

The turbulent boundary layer is visualized, on the basis of reported experimental measurements, as consisting of three zones—a laminar sublayer adjacent to the wall, a buffer zone and the turbulent zone. Change in the velocity from zero to V is virtually confined within the first two zones.

4.1.2. Thermal Boundary Layer

Heat transfer from a hot plate to a flowing fluid occurs principally by convection. Similar to the velocity boundary layer, a thermal boundary layer is formed in the liquid that includes laminar, transition and turbulent zones in the sense of heat transport depending upon the variation of the Reynolds number along the plate. The thickness of the thermal boundary layer is, however, different from that of the hydrodynamic boundary layer, and considerably depends upon the thermal properties of the fluid. Just at the surface of the plate, the temperature of the fluid will be the same as the temperature of the plate (continuity of temperature). Again, the

fluid temperature will be very close to the bulk temperature at the edge of the thermal boundary layer (i.e. at the edge of the thermal boundary layer, the difference from the bulk temperature remains within 1% of the total temperature drop across the boundary layer). Thus, it is not through the so-called stagnant film, but through the boundary layer that convective transfer of heat from a surface occurs.

Velocity distributions in a two-dimensional boundary layer [i.e. how the x - and the y -components of the velocity, u and v , depend upon the position (x, y)] govern the heat transfer coefficient in the boundary layer. A correlation that can be used to calculate the heat transfer coefficient (which is a function of x) will be described later. The thermal boundary layer over a flat plate is shown in Fig. 4.1(b). It may be noted that when the dimensionless group called the Prandtl number, $Pr = \nu/\alpha$ ($\nu = \mu/\rho$ is called the *momentum diffusivity*, and $\alpha = k/\rho c_p$ is called the *thermal diffusivity*) is less than unity, the momentum boundary layer remains within the thermal boundary layer. If $Pr > 1$, the reverse becomes true.

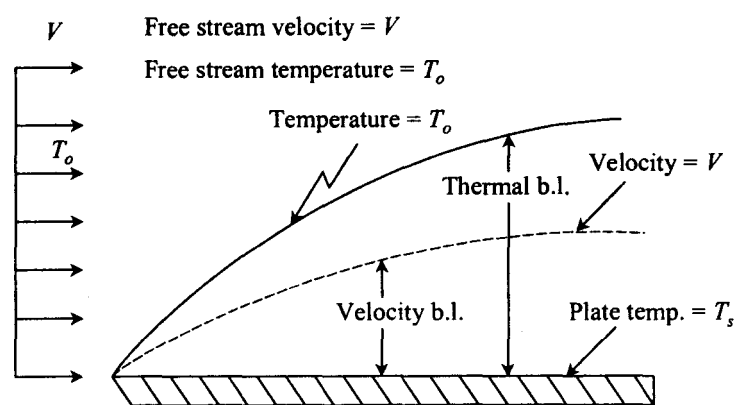


Fig. 4.1(b) Laminar velocity and temperature boundary layers over a heated plate.

4.1.3. Flow across a Cylinder

Transverse flow of a fluid across a single cylinder or a bank of cylinders is an important flow situation in industrial heat transfer. For example, a hot process stream flowing through a pipe may be cooled by blowing air across the pipe. Here too, a boundary layer forms over the cylinder as shown in Fig. 4.2. The point A is called the *stagnation point* because the fluid velocity is zero at this point and the pressure is maximum (the physical reason being conversion of velocity head to pressure head). The distance of a point on the cylinder from the stagnation point is given by the linear measure x or the angular measure θ . As one moves away from the stagnation point along the periphery, the boundary layer decelerates and its thickness increases. The velocity profile within the boundary layer becomes gradually more and more flatter. Farther away from the stagnation point, the retarded layer is unable to overcome the static pressure and the boundary layer 'separates' from the cylinder surface. Downstream of this point of separation, reverse flow sets in near the surface but forward flow continues in the outer part of the boundary layer. On the rear side of the cylinder there are wakes, and the flow becomes turbulent.

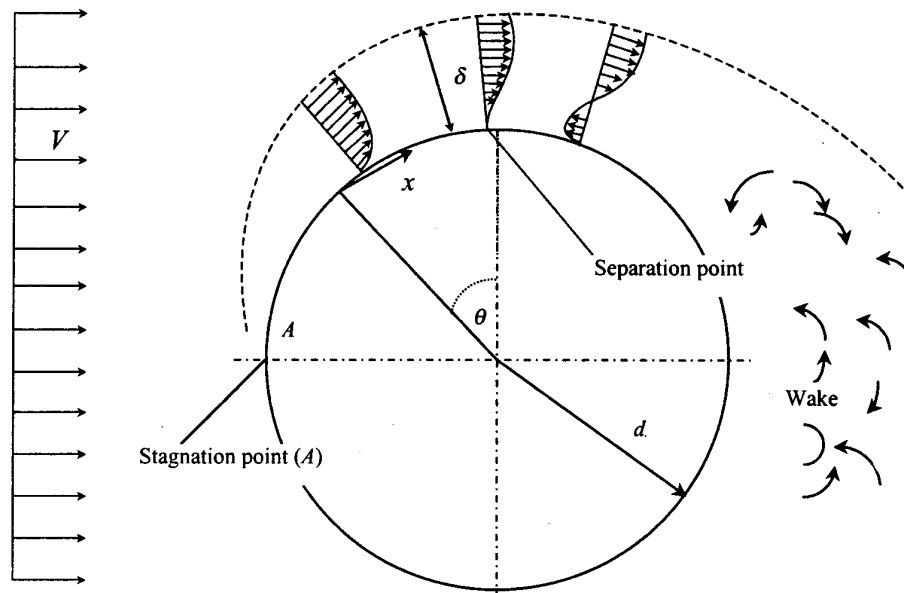


Fig. 4.2 Velocity profile for cross-flow over a cylinder.

For practical heat transfer calculations, an 'average heat transfer coefficient' rather than the local coefficient is used. We will now discuss the methods and the theoretical basis of development of correlations for heat transfer coefficients in forced convection.

4.2. DIMENSIONLESS GROUPS IN HEAT TRANSFER

Before we discuss the important heat transfer correlations available in the literature, it is pertinent to make a list and also to mention the physical significances of the dimensionless groups which are frequently used in forced convection heat transfer (Table 4.2).

Table 4.2 Important dimensionless groups in convective heat transfer

Name	Expression	Physical significance
Nusselt number, $Nu = \frac{hl}{k}$		$\frac{\text{Wall temperature gradient}}{\text{Temperature gradient across the fluid in the pipe}}$
Reynolds number, $Re = \frac{IV\rho}{\mu}$		$\frac{\text{Inertial force}}{\text{Viscous force}}$
Prandtl number, $Pr = \frac{c_p\mu}{k} = \frac{\nu}{\alpha}$		$\frac{\text{Momentum diffusivity}}{\text{Thermal diffusivity}}$
Stanton number, $St = \frac{h}{V\rho c_p}$		
	$= \frac{(hl/k)}{(IV\rho/\mu)(c_p\mu/k)} = \frac{Nu}{(Re)(Pr)}$	$\frac{\text{Rate of wall heat transfer by convection}}{\text{Rate of heat transfer by bulk flow}}$
Peclet number, $Pe = \frac{\rho V l c_p}{k} = \frac{IV\rho}{\mu} \cdot \frac{c_p\mu}{k}$ $= Re \cdot Pr$		$\frac{\text{Rate of heat transfer by bulk flow}}{\text{Rate of heat transfer by conduction}}$
Graetz number, $Gz = Pe \cdot d/L = Re \cdot Pr \cdot d/L$		Similar to Peclet number (used in connection with analysis of heat transfer in laminar flow in pipes)

The above table needs some more explanation. In these expressions (except the last one, Gz), l denotes the *characteristic length*. A characteristic length can be defined based on the geometry of the system. For example, in the case of flow, heat or mass transfer in a pipe, the pipe diameter is taken as the characteristic length. For flow past a flat plate, the distance from the leading edge is the characteristic length. So, in the case of pipe flow we use d as the characteristic length in place of l . In the expression for Graetz number, L is the pipe length over which heat transfer occurs.

4.3. CORRELATIONS FOR THE HEAT TRANSFER COEFFICIENT-INTERNAL FLOWS

Systems of various shapes and different flow conditions are encountered in practical heat transfer calculations. The common shapes or geometries are cylindrical, spherical or flat (or planar). The most common flow problem involves flow through a circular pipe; the other important ones are flow through a non-circular duct (e.g. a rectangular channel), through a packed bed (e.g. of catalysts or of some inert material forming a regenerative bed), in an agitated vessel with internals, through the jacket over a reaction kettle or vessel, across a bank of tubes, etc. Again, the flow of a fluid in a system may be internal (e.g. flow through a pipe) or external (e.g. flow across the outer surface of a pipe). Here, we describe the more important correlations along with the ranges of applicability for the calculation of heat transfer coefficients. Some of these were developed even more than half a century ago, but are still widely used in engineering calculations. The important correlations for the case of internal flow are discussed below.

4.3.1. Laminar Flow through a Circular Pipe

Laminar flow ($Re < 2100$) may occur in industrial equipment for heating or cooling of a considerably viscous liquid or of a liquid or solution sensitive to shear stress. Because it is possible to analytically solve the problem of heat transfer to or from a fluid in laminar flow through a pipe, a large volume of theoretical research work has been published in this area. Details are available in Shah and London (1978). For fully developed flow and a constant temperature of the pipe wall, the following correlation of Hausen (1943), although outdated, is recommended. This correlation incorporates a term called the 'thermal entry length'. (It is the distance from the point of entry to a pipe or channel over which the 'boundary layer approximations' are applicable; this will be discussed later.)

$$Nu = 3.66 + \frac{0.0668(d/L)(Re)(Pr)}{1 + 0.04[(d/L)(Re)(Pr)]^{2/3}} \quad \left[Nu = \frac{\bar{h}d}{k}, Re = \frac{dV\rho}{\mu} \right] \quad (4.1)$$

where

L is the pipe length

d is the pipe diameter

h is the average heat transfer coefficient over the pipe length L including the entry length.

A simpler correlation was suggested by Sieder and Tate (1936).

$$Nu = 1.86 \left[\frac{(Re)(Pr)}{L/d} \right]^{1/3} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (4.2)$$

The above correlation, Eq. (4.2), is applicable when

- (a) $0.48 < Pr < 16700$,
- (b) the viscosity ratio is within the range, $0.0044 < \mu/\mu_w < 9.75$, where μ is the viscosity of the fluid at the bulk temperature, and μ_w is that at the wall temperature, T_w
- (c) $(Pe \, d/L) > 10^8$.

The fluid properties have to be known or evaluated at the mean bulk temperature of the fluid.

4.3.2. Turbulent Flow through a Circular Pipe

For ‘fully developed’ turbulent flow through a pipe, the following correlation suggested by Dittus and Boelter (1930) is used widely.

$$Nu = 0.023 (Re)^{0.8} (Pr)^n \quad (4.3)$$

where

- $n = 0.4$ for heating ($T_w > T$)
- $n = 0.3$ for cooling ($T_w < T$).

The conditions for applicability of this correlation are: (a) $0.7 \leq Pr \leq 160$; (b) $d/L < 0.1$; and (c) $Re \geq 10\,000$.

The above correlation, Eq. (4.3), is applicable for moderate values of the temperature difference between the wall and the bulk, $T_w - T$. The fluid properties are evaluated at the arithmetic mean of the bulk temperatures (i.e. the average of the inlet and the outlet temperatures of the fluid). The maximum error in the predicted value of the heat transfer coefficient does not exceed 25%.

When the temperature difference between the wall and the bulk is substantial, its effect on the fluid properties, particularly on viscosity, needs to be taken into account. For such cases, the Sieder-Tate equation (1936) is recommended.

$$Nu = 0.027 (Re)^{0.8} (Pr)^{0.33} (\mu/\mu_w)^{0.14} \quad (4.4)$$

Here, μ/μ_w is the viscosity correction factor that has to be used when the wall viscosity at the wall temperature, μ_w , is substantially different from that, μ , at the bulk temperature. The conditions of applicability are: (a) $0.7 \leq Pr \leq 16700$; (b) $Re \geq 10\,000$; and (c) $d/L \leq 0.1$. The bulk fluid properties have to be evaluated at the arithmetic mean bulk temperature.

Whitaker (1972) suggested the following equation which is based on the experimental data of a large number of researchers.

$$Nu = 0.015 (Re)^{0.83} (Pr)^{0.42} (\mu/\mu_w)^{0.14} \quad (4.5)$$

In the case of heat transfer to or from a liquid metal, flowing through a pipe, at a constant wall temperature, the correlation of Seban and Shimazaki (1951) may be used.

$$Nu = 5.0 + 0.025 (Pe)^{0.8} \quad (4.6)$$

Equation (4.6) is applicable when $Pe > 100$ and $L/d > 60$. All the properties of the liquid metal should be evaluated at the mean bulk temperature.

4.3.3. Flow through a Non-circular Duct

Ducts of non-circular cross-sections, rectangular or square, are often used in the industrial and other applications. Flow through the annulus of a double-pipe heat exchanger, flow through the shell side of a shell-and-tube heat exchanger and flow of hot exhaust gases through rectangular or square ducts are a few common examples. The foregoing equations for estimation of the Nusselt number are generally applicable for heat transfer calculations for flow through non-circular ducts, but the *equivalent diameter* of the duct is to be used in the calculation of the Reynolds number. The equivalent diameter d_e is four times the *hydraulic radius*, r_H . The hydraulic radius is defined as

$$r_H = \frac{\text{Cross-sectional area of flow}}{\text{Wetted perimeter}}$$

and

$$d_e = 4r_H \quad (4.7a)$$

For example, in the case of flow through a rectangular duct of sides l_1 and l_2 the equivalent diameter is

$$d_e = \frac{4l_1l_2}{2(l_1 + l_2)}$$

It should be noted that the calculation of the wetted perimeter of a duct in the case of heat transfer calculation may be different from that used in the case of determination of pressure drop (Kem, 1950; Ludwig, 1983). To illustrate this point, we consider the flow through the annulus between two concentric tubes in a double-pipe heat exchanger (the construction of this kind of heat exchanger will be described later). Let the outside diameter of the inner pipe be d_1 , and the inside diameter of the outer pipe be d_2 . If our objective is to calculate the pressure drop for flow through this annulus, the wetted perimeter is $\pi(d_1 + d_2)$, because both the walls forming the annulus contribute to the frictional pressure drop. But if we consider heat transfer from a hot fluid flowing through the inner pipe to a cold fluid flowing through the annulus, *wetting* of the outer wall of the inner pipe only becomes relevant to the heat transfer coefficient in the annulus. So, here the wetted perimeter is simply πd_1 . The equivalent diameters of the annulus in the respective cases are:

For pressure drop calculation,

$$d_{e,f} = 4 \frac{(\pi/4)(d_2^2 - d_1^2)}{\pi(d_1 + d_2)} = d_2 - d_1 \quad (4.7b)$$

For heat transfer calculation,

$$d_{e,h} = 4 \frac{(\pi/4)(d_2^2 - d_1^2)}{\pi d_1} = \frac{1}{d_1}(d_2^2 - d_1^2) \quad (4.7c)$$

4.4. CORRELATIONS FOR THE HEAT TRANSFER COEFFICIENT-EXTERNAL FLOWS

Convective heat transfer in external flows is encountered in many practical situations. Typical cases are flow over a flat surface, along or across a single pipe or a pipe bundle, over a sphere or a bed of spheres (e.g. convective heat transport from a wall, heating of air by a bundle of steam

pipes, heat transfer in a packed bed of catalyst or that in a fluidized bed, etc.). Several important correlations that cover more common cases of external flows are discussed below.

4.4.1. Flow over a Flat Plate

Heat transfer in flow over a flat plate occurs through the boundary layer formed on the plate. A physical description of the boundary layer including the laminar, transition and turbulent zones has been given in Section 4.1. The following correlations are suggested for heat transfer calculations in the boundary layer flow.

For *heat transfer in laminar boundary layer flow*, the following correlation for the local Nusselt number can be obtained from an approximate solution of the boundary equations of momentum and energy. It depends upon the distance from the leading edge of the plate.

$$Nu_x = 0.332 (Re_x)^{1/2} (Pr)^{1/3} \quad (0.5 \leq Pr \leq 50) \quad (4.8a)$$

where

Nu_x is the local Nusselt number $= xh_x/k$

Re_x is the local Reynolds number $= xV\rho/\mu$

Here x is the distance from the leading edge of the plate, and h_x is the local heat transfer coefficient. An average value of the heat transfer coefficient over a distance L may be obtained as

$$h_{av} = \frac{1}{L} \int_0^L h_x dx$$

or

$$Nu_L = 0.664 (Re_L)^{1/2} (Pr)^{1/3} \quad (4.8b)$$

For *heat transfer in turbulent boundary layer flow*, a simple correlation in common use is

$$Nu_x = 0.0296 (Re_x)^{4/5} (Pr)^{1/3}; \quad 5 \times 10^5 < Re_x < 10^7, \quad 0.6 < Pr < 60 \quad (4.9)$$

4.4.2. Flow across a Cylinder

Quite a few correlations for the average Nusselt number are available for the case of convective heat transfer in flow across a cylinder. For cross flow of a liquid over a cylinder, the heat transfer coefficient can be calculated by using:

(a) The correlation proposed by Fand (1965)

$$Nu = \frac{hd}{k} = [0.35 + 0.56(Re)^{0.52}] (Pr)^{0.3}; \quad Re = \frac{dV\rho}{\mu}; \quad 0.1 < Re < 10^5 \quad (4.10)$$

(b) The correlations of Eckert and Drake (1972)

$$Nu = [0.43 + 0.50(Re)^{0.5}] (Pr)^{0.38} (Pr/Pr_w)^{0.25}; \quad 1 < Re < 10^3 \quad (4.11)$$

$$Nu = 0.25(Re)^{0.6} (Pr)^{0.38} (Pr/Pr_w)^{0.25}; \quad 10^3 < Re < 2 \times 10^5 \quad (4.12)$$

where Pr_w is the Prandtl number at the wall temperature. These correlations are applicable to both gases and liquids.

(c) The correlation of Churchill and Burnstein (1977)

$$\text{Nu} = 0.3 + \frac{0.62(\text{Re})^{1/2}(\text{Pr})^{1/3}}{\left[1 + (0.4/\text{Pr})^{2/3}\right]^{1/4}} \left[1 + \left(\frac{\text{Re}}{2.82 \times 10^5}\right)^{5/8}\right]^{4/5} \quad (4.13)$$

for $10^2 < \text{Re} < 10^7$; $(\text{Re})(\text{Pr}) > 0.2$

The above correlation is based on the data for flow of air, water and liquid sodium. In the intermediate range of Reynolds number, however, the following correlation predicts the heat transfer coefficient more accurately.

$$\text{Nu} = 0.3 + \frac{0.62(\text{Re})^{1/2}(\text{Pr})^{1/3}}{\left[1 + (0.4/\text{Pr})^{2/3}\right]^{1/4}} \left[1 + \left(\frac{\text{Re}}{2.82 \times 10^5}\right)^{1/2}\right] \quad (4.14)$$

for $2 \times 10^4 < \text{Re} < 4 \times 10^5$; $(\text{Re})(\text{Pr}) > 0.2$

(d) Whitaker (1972) suggested the following correlation that involves a maximum error of $\pm 25\%$.

$$\text{Nu} = [0.40 (\text{Re})^{1/2} + 0.06 (\text{Re})^{2/3}] (\text{Pr})^{0.4} (\mu/\mu_w)^{0.25} \quad (4.15)$$

for $40 < \text{Re} < 105$, $0.65 < \text{Pr} < 300$ and $0.25 < \mu/\mu_w < 5.2$

4.4.3. Flow past a Sphere

The following important correlations are available for the case of heat transfer in flow over a single sphere:

(a) For flow of liquids past a sphere, Kramers (1946) proposed the following correlation:

$$\text{Nu} (\text{Pr})^{-0.3} = 0.97 + 0.68 (\text{Re})^{0.5}; \quad 1 < \text{Re} < 2000 \quad (4.16)$$

(b) Whitaker (1972) suggested the following correlation:

$$\text{Nu} = 2 + [0.4 (\text{Re})^{1/2} + 0.06 (\text{Re})^{2/3}] (\text{Pr})^{0.4} (\mu/\mu_w)^{0.25} \quad (4.17)$$

where μ is the viscosity of the fluid at the bulk temperature, and μ_w is that at the wall temperature. This correlation is applicable to both gases and liquids, and the error in the prediction remains within $\pm 30\%$. The number '2' arises out of the contribution of conduction only when there is no motion in the medium, i.e. when $\text{Re} = 0$. The Reynolds number is based on the diameter of the sphere.

4.5. HEAT TRANSFER WITH A VARIABLE DRIVING FORCE-COCURRENT AND COUNTERCURRENT OPERATIONS

The calculation of the rate of heat transfer becomes very simple if the overall heat transfer coefficient, the area of heat transfer and the temperature driving force all remain constant. But in many cases, one or more of these quantities may vary. Examples of heat transport through a variable area (in the case of radial transport through a cylinder or a sphere) or with a variable heat transfer coefficient (in the case of boundary layer heat transfer) have been discussed. However, variation of the temperature driving force with position in a heat transfer equipment or device is more common. In order to explain this, we will consider a very simple type of heat transfer device-the double-pipe heat exchanger.

The schematic of a double-pipe heat exchanger is shown in Fig. 4.5 (cocurrent) and Fig. 4.6 (countercurrent). The device consists of two concentric pipes welded at the ends and provided with flow nozzles so that a hot and a cold fluid can be brought in thermal contact in order to effect transfer of heat. One of the fluids flows through the pipe and the other through the annulus. If both the streams, the hot and the cold, flow in the same direction, the flow arrangement is called *parallel* or *cocurrent*. If the hot and the cold streams flow in the opposite directions, the arrangement is called *countercurrent* or *counterflow*. The countercurrent flow arrangement is more common because it allows exchange of more amount of heat between the streams than that by the cocurrent configuration.

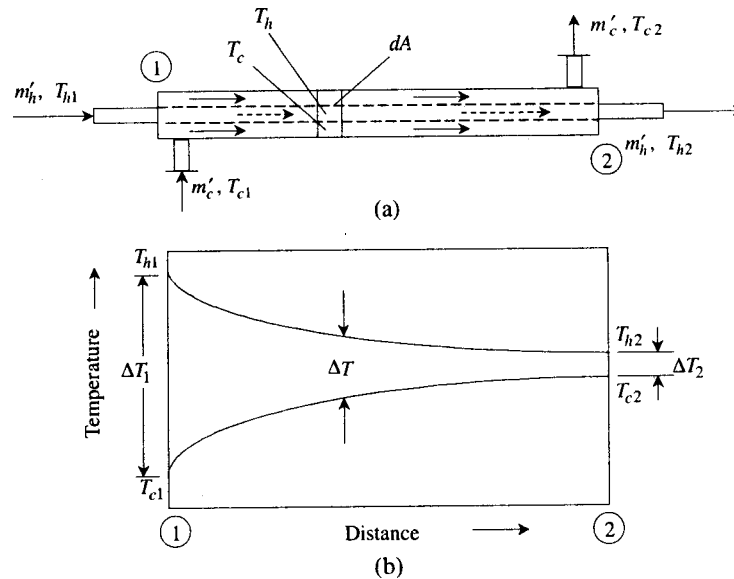


Fig. 4.5 (a) Double-pipe heat exchanger in parallel or cocurrent flow and (b) temperature distribution.

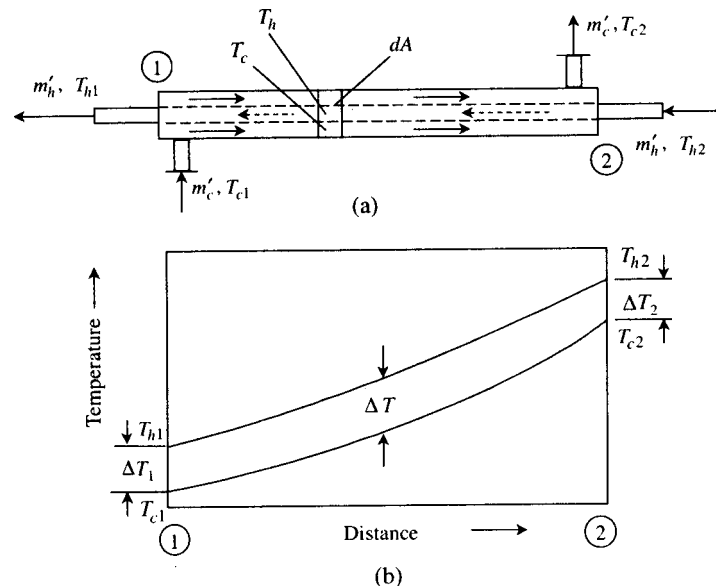


Fig. 4.6 (a) Double-pipe heat exchanger in countercurrent flow and (b) temperature distribution.

Let us consider the schematic of a cocurrent flow system shown in Fig. 4.5. The following notations will be used:

m' is the flow rate of the fluid stream, kg/s
 c_p is the specific heat, J/kg °C

T_1, T_2 are the temperatures of the fluid stream at the ends 1 and 2 of the device, respectively

$\Delta T = T_h - T_c$ = the driving force at any section.

The subscript 'h' refer to the hot fluid and 'c' to the cold fluid, and T_h and T_c are the 'local temperatures' of the respective streams. We arbitrarily assume that the hot fluid flows through the inner pipe and transfers heat through its wall to the cold fluid flowing through the annulus as shown in the figure. The temperatures of both the streams vary with the position along the device.

Now consider a 'thin' section of the device having a heat transfer area dA (i.e. the area of the wall of the inner pipe in this thin section is dA). The local temperatures of the fluids are T_h and T_c for the hot and the cold fluid, respectively. If U is the overall heat transfer coefficient (assumed constant), the rate of heat transfer, dQ , at steady state through the small area dA may be written as

$$dQ = U dA (T_h - T_c) = U dA \Delta T; \quad \Delta T = (T_h - T_c) \quad (4.24)$$

Also, a heat balance over the thin section gives

$$dQ = -m'_h c_{ph} dT_h = m'_c c_{pc} dT_c \quad (4.25)$$

Here, dT_h and dT_c are the changes in the temperatures of the hot and the cold streams over the thin section because of exchange of heat. The hot stream cools down as it flows through the pipe and hence dT_h is negative. So a negative sign is incorporated in the term in the middle of Eq. (4.25) to make it consistent with respect to sign.

Next we put $T_h = \Delta T + T_c$ in Eq. (4.25) and eliminate T_c . Thus

$$dQ = -m'_h c_{ph} d(\Delta T + T_c) = -m'_h c_{ph} d(\Delta T) - m'_h c_{ph} dT_c = -m'_h c_{ph} d(\Delta T) - \frac{m'_h c_{ph}}{m'_c c_{pc}} dQ$$

Therefore,

$$dQ = -\frac{m'_h c_{ph} m'_c c_{pc}}{m'_h c_{ph} + m'_c c_{pc}} d(\Delta T) \quad (4.26)$$

Substituting for dQ from Eq. (4.26) in Eq. (4.24) and arranging, we get

$$\frac{d(\Delta T)}{\Delta T} = -\frac{m'_h c_{ph} + m'_c c_{pc}}{m'_h c_{ph} m'_c c_{pc}} U dA \quad (4.27)$$

We now integrate Eq. (4.27) from end 1 to end 2 of the system

At end 1, $A = 0$ and $\Delta T = T_{h1} - T_{c1} = \Delta T_1$ (say)

At end 2, $A = A$ and $\Delta T = T_{h2} - T_{c2} = \Delta T_2$ (say).

Therefore,

$$\begin{aligned}
- \int_{\Delta T_1}^{\Delta T_2} \frac{d(\Delta T)}{\Delta T} &= \frac{m'_h c_{ph} + m'_c c_{pc}}{m'_h c_{ph} m'_c c_{pc}} U \int_0^A dA \\
\ln \frac{\Delta T_1}{\Delta T_2} &= \frac{m'_h c_{ph} + m'_c c_{pc}}{m'_h c_{ph} m'_c c_{pc}} UA
\end{aligned} \tag{4.28}$$

Now we write the heat balance equations for both the streams, the hot stream and the cold stream, over the heat exchange device. If Q is the total rate of heat transfer, we have

$$Q = m'_h c_{ph} (T_{h1} - T_{h2}) = m'_c c_{pc} (T_{c2} - T_{c1}) \tag{4.29}$$

or

$$\begin{aligned}
\frac{Q}{m'_h c_{ph}} + \frac{Q}{m'_c c_{pc}} &= (T_{h1} - T_{h2}) + (T_{c2} - T_{c1}) \\
&= (T_{h1} - T_{c1}) - (T_{h2} - T_{c2}) = \Delta T_1 - \Delta T_2
\end{aligned}$$

Therefore,

$$Q = \frac{m'_h c_{ph} m'_c c_{pc}}{m'_h c_{ph} + m'_c c_{pc}} (\Delta T_1 - \Delta T_2) \tag{4.30}$$

If we define ΔT_m as an appropriate *mean driving force* over the entire length of the device, we have

$$Q = U A \Delta T_m \tag{4.31}$$

From Eqs. (4.30) and (4.31), we get

$$UA \Delta T_m = \frac{m'_h c_{ph} m'_c c_{pc}}{m'_h c_{ph} + m'_c c_{pc}} (\Delta T_1 - \Delta T_2) \tag{4.32}$$

Comparing Eqs. (4.28) and (4.32), we get

$$\Delta T_m = \frac{\Delta T_1 - \Delta T_2}{\ln (\Delta T_1 / \Delta T_2)} \tag{4.33}$$

Equation (4.33) is one of the most important equations in heat transfer calculation. It shows that when the temperature driving force varies from one end of a heat exchange device to the other, the *log mean temperature difference*, ΔT_m , given above is the applicable mean driving force.

Next we will consider the case of heat exchange between a hot and a cold fluid in countercurrent flow as shown in Fig. 4.6. The notations used are very much the same as those used in the case of cocurrent flow.

The rate of heat transfer at steady state through the elementary area dA is

$$dQ = U dA (T_h - T_c) = U dA \Delta T \tag{4.34}$$

Also, if dT_h and dT_c are the temperature changes of the streams over this elementary area, we have

$$dQ = - m'_h c_{ph} dT_h = m'_c c_{pc} dT_c \tag{4.35}$$

The heat balance equations for the countercurrent flow situation are similar to those for the cocurrent flow. It is easy to verify that the final form of the mean driving force remains the same as Eq. (4.36) and the rate of heat transfer is then given by Eq. (4.31). The log mean driving force is abbreviated as LMTD. This is left as an exercise for the students.

PROBLEMS

- Nitrogen gas is heated at a rate of 2000 kg/h before passing it through the trays in a tray drier in which an organic product is being dried. Medium pressure steam available from a waste heat boiler at a pressure of 5.7 bar (saturation temperature = 160°C) condenses within the tubes of a 'finned tube' heater. The gas flows outside the tubes. The heat duty is 38,700 kcal/h. The specific heat of nitrogen can be taken as 0.239 kcal/kg °C. The heat exchanger has an area of 10 m² and the overall heat transfer coefficient is estimated at 70 kcal/h m² °C. The fin efficiency is 63%. Calculate the inlet and the outlet temperatures of the gas.
- Hot water is flowing through a 3.5 cm (1-1/4") schedule 40 steel pipe at a velocity of 1.8 m/s. The inlet temperature is 110°C, and the length of the pipe is 15 m. A 2 cm thick layer of insulation ($k_c = 0.12$ W/m °C) covers the pipe. The outside film coefficient is 10 W/m² °C, and the ambient temperature is 20°C. Calculate the drop in the temperature of the water over this section of the pipe.

Properties of water at 110°C: $\rho = 950$ kg/m³; $c_p = 4.23$ kJ/kg °C; $\mu = 2.55 \times 10^{-4}$ kg/m s; $k = 0.685$ W/m °C.

Actual i.d. of the pipe, $d_i = 3.5$ cm, $r_i = 0.0175$ m; wall thickness of 1-1/4 schedule 40 pipe = 0.0036 m; outer radius of the pipe, $r' = 0.0175 + 0.0036 = 0.0211$; outer radius of the insulation, $r_o = 0.0211$ m + 0.02 m = 0.0411 m; pipe length = 15 m; water velocity, $V = 1.8$ m/s; $h_o = 10$ W/m² °C; thermal conductivity of the pipe wall (steel), $k_w = 43$ W/m °C; thermal conductivity of the insulation, $k_c = 0.12$ W/m °C.
- A 41 mm i.d. (1-1/2" nominal bore) schedule 40 pipe carries water flowing at a rate of 1 kg/s. Water enters the pipe at 28°C and is heated by a stream of hot flue gas in cross flow over the pipe. The gas velocity is 10 m/s. The arrangement essentially aims at recovering a part of the waste heat of the gas stream which has a bulk temperature of 250°C. The pressure is essentially atmospheric. The length of the pipe is 20 m. At what temperature does the water leave the pipe? The properties of the flue gas are about the same as those of the air.

Data: For water at 28°C: $\rho = 996$ kg/m³; $c_p = 1.0$ kcal/kg °C; $\mu = 8.6 \times 10^{-4}$ kg/m s; $k_w = 0.528$ kcal/h m °C.; flow rate = 1 kg/s = 3600 kg/h.

For the flue gas (same as air): $\rho = 0.891$ kg/m³; $c_p = 0.243$ kcal/kg °C; $\mu = 2.33 \times 10^{-5}$ kg/m s; thermal conductivity, $k_a = 0.0292$ kcal/h m °C; Pr = 0.69.

Inner diameter of the tube, $d_i = 41$ mm = 4.1×10^{-2} m
- Lubricating oil used in the gearbox of a 14,000 rpm high speed blower is being recycled continuously through a double-pipe counterflow heat exchanger for cooling. The oil is to be cooled from 70°C to 40°C at the rate of 1000 kg/h using water entering at 28°C. The water temperature at the exit should not exceed 42°C. The specific heat of oil is 2.05 kJ/kg °C and that of water is 4.17 kJ/kg °C. Calculate the required rate of flow of water. If the heat exchange area is 3.0 m², calculate the overall heat transfer coefficient.

